EASY ALGEBRA

Linear functions

06

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ALGEBRA

In science class, Diego conducted an experiment to learn about evaporation. Diego measured the height of fluid in a beaker over a period of time. The function $f x = 39 - 0.18x$ gives the estimated height, in centimeters cm, of the fluid in the beaker x days after the start of the experiment. Which of the following is the best interpretation of 39 in this context?

A. The estimated height, in cm, of the fluid at the start of the experiment

B. The estimated height, in cm, of the fluid at the end of the experiment

C. The estimated change in the height, in cm, of the fluid each day

D. The estimated number of days for all the fluid to evaporate

$$gx = 11x + 4$$

For the given linear function g, which table shows three values of x and their corresponding values of gx?

A.

x	gx
-1	7
0	11
1	15

B.

x	gx
-1	-4
0	0
1	4

C.

x	gx
-1	-7
0	4
1	15

D.

x	gx
-1	-11
0	0
1	11

Q3 ALGEBRA

The function g is defined as $g(x) = 5x + a$, where a is a constant. If $g(4) = 31$, what is the value of a ?

A. 30

B. 22

C. 11

D. -23

Q4

ALGEBRA

The function f is defined by $f x = - 3x + 60$. What is the value of $f x$ when $x = - 8$?

A. 49

B. 52

C. 57

D. 84

Q5 ALGEBRA

For a training program, Juan rides his bike at an average rate of 5.7 minutes per mile. Which function m models the number of minutes it will take Juan to ride x miles at this rate?

A. $mx = \frac{x}{5.7}$

B. $mx = x + 5.7$

C. $mx = x$ - 5.7

D. $mx = 5.7x$

Q6

GRID-IN ALGEBRA

If $y = 5x + 10$, what is the value of y when $x = 8$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7 ALGEBRA

The function g is defined by $gx = 10x + 8$. What is the value of gx when $x = 8$?

A. 0

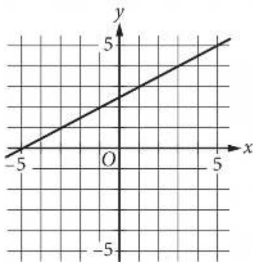
B. 8

C. 10

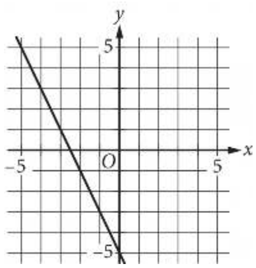
D. 88

Which of the following is the graph of the equation $y = 2x - 5$ in the xy-plane?

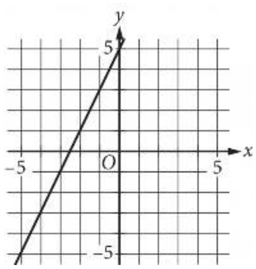
A.



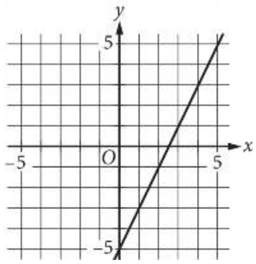
B.



C.



D.


y
5
x
−5
O
5
−5

Q9 ALGEBRA

x	$f(x)$
0	29
1	32
2	35

For the linear function f, the table shows three values of x and their corresponding values of $f(x)$. Which equation defines $f(x)$?

A. $f(x) = 3x + 29$

B. $f(x) = 29x + 32$

C. $f(x) = 35x + 29$

D. $f(x) = 32x + 35$

Q10 ALGEBRA

The function f is defined by $f x = \frac{1}{10}x$ - 2. What is the y-intercept of the graph of $y = f x$ in the xy-plane?

A. -2, 0

B. 0, -2

C. 0, $\frac{1}{10}$

D. $\frac{1}{10}$, 0